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BetterCallMain

These are the parts from my code that includes any use of AI:

Decks.java

```
// Pick random player
Random rand = new Random();
String firstPlayer = rand.nextBoolean() ? config.player1Name() :
config.player2Name();
```

GameEngineImpl.java

```
// UI and Network listeners
private final List<GameEventListener> listeners = new ArrayList<>();
```

```
public boolean canFlip(int cardId) {
    if (cardId < 0 || cardId >= board.size() || gameOver) return false;
    Card card = board.get(cardId);
    return !card.removed() && !card.faceUp();
}
```

```
// Notify listeners that a card was flipped
GameSnapshot snap = getSnapshot();
for (GameEventListener listener : listeners) {
    listener.onCardFlipped(cardId, snap);
}
```

```
List<Integer> idsToResolve = List.copyOf(flippedIds);
```

```
flippedIds.clear();
```

```
// Notify listeners of the mismatch before flipping cards down
snap = getSnapshot();
for (GameEventListener listener : listeners) {
    listener.onMismatch(idsToResolve, snap);
}
```

```
snap = getSnapshot();
for (GameEventListener listener : listeners) {
    listener.onMismatchFlipback(idsToResolve, snap);
    listener.onTurnChanged(activePlayer, snap);
}
```

```
if (!gameOver) throw new IllegalStateException("Game is not over yet.");
```

```

@Override
public void removeListener(GameEventListener listener) {
    listeners.remove(listener);
}

```

SymbolLoader.java

```

public class SymbolLoader {

    private static final String SYMBOL_PATH =
"src/main/resources/Images/Memory";
    private static final String CARD_BACK_PATH = SYMBOL_PATH +
"/backface.png";
    private static final Map<Integer, BufferedImage> symbolCache = new
HashMap<>();
    private static BufferedImage cardBackCache = null;

    public static BufferedImage getSymbolImage(int symbolId) {
        if (symbolCache.containsKey(symbolId)) {
            return symbolCache.get(symbolId);
        }

        String resourceName = SYMBOL_PATH + symbolId + ".png";
        BufferedImage img = loadImage(resourceName);
        if (img != null) {
            symbolCache.put(symbolId, img);
        }
        return img;
    }

    public static BufferedImage getCardBackImage() {
        if (cardBackCache == null) {
            cardBackCache = loadImage(CARD_BACK_PATH);
        }
        return cardBackCache;
    }

    private static BufferedImage loadImage(String path) {
        try (InputStream is = SymbolLoader.class.getResourceAsStream(path))
        {
            if (is == null) {
                System.err.println("Resource not found: " + path);
                return null;
            }
            return ImageIO.read(is);
        } catch (IOException e) {
            System.err.println("Error loading image at " + path + ": " +
e.getMessage());
            return null;
        }
    }
}

```